



HAMMARD UNIVERSITY BANGLADESH

Department of Electrical and Electronics Engineering
Faculty of Science, Engineering and Technology

Course Code: EEE 117

Course Title: Electrical Engineering

Final Examination

Semester: Spring 2026

Date: 9/15/2026

Total Time: 2:00 Hours

Total Marks: 40

Answer any Four (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- | | CLOs | Marks |
|--|------|-------|
| Q1. (a) Define capacitor, capacitance, inductor and inductance. | CLO1 | [2] |
| (b) Determine the current through a capacitor whose voltage is shown in Fig. 1(b). | CLO3 | [4] |

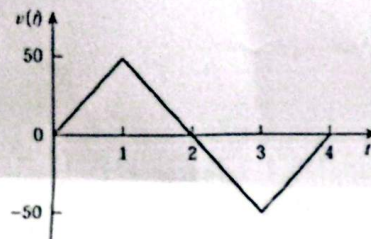


Fig. 1(b)

- | | | |
|---|------|-----|
| (c) Derive the energy equation for both inductor and capacitor. | CLO2 | [4] |
|---|------|-----|
- Q2. (a) Define Natural response and time constant. CLO1 [2]
- (b) Derive the source free RC circuit response with appropriate graphs for voltage. CLO2 [4]
- (c) The switch in the circuit of Fig. 2(c) has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$. CLO3 [4]

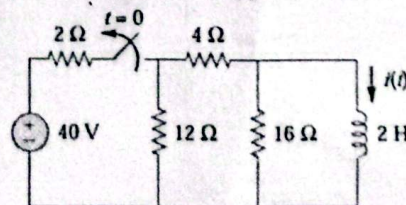


Fig. 2(c)

- | | | |
|---|------|-----|
| Q3. (a) Define Sinusoid, Admittance and Phasor. | CLO1 | [2] |
| (b) Using the phasor approach, | CLO3 | [4] |

- i. Determine the current in a circuit described by the integrodifferential equation:

$$4i + 8 \int i dt - 3 \frac{di}{dt} = 50 \cos(2t + 75^\circ)$$

- ii. Determine the voltage in a circuit described by the integrodifferential equation

$$2 \frac{dv}{dt} + 5v + 10 \int v dt = 50 \cos(5t - 30^\circ)$$

- (c) Find the value of $v_0(t)$ and current through 60Ω resistor for the circuit in Fig. 3(c). CLO3 [4]

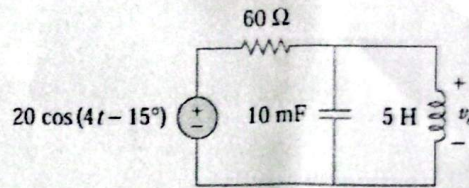


Fig. 3(c)

- Q4. (a) Using Nodal analysis find the value of i_x and verify whether the current flowing through $1H$ and $0.5H$ are same or not. CLO3 [8]

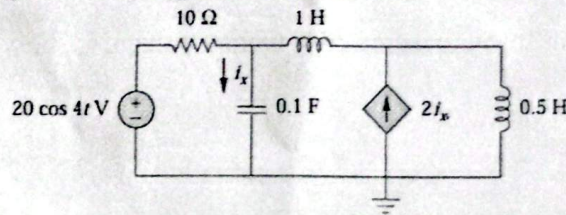


Fig. 4(a)

- (b) Describe the reasons for applying supermesh and supernode process while using mesh and nodal analysis respectively. CLO2 [2]

- Q5. (a) Write a short note on the Maximum Power Transfer theorem. CLO1 [2]

- (b) Derive the equation of maximum power transfer theorem and note down the formulas. CLO2 [3]

- (c) Find the value of maximum power transfer in the circuit of Fig. 5(c) and also find the maximum power. CLO3 [5]

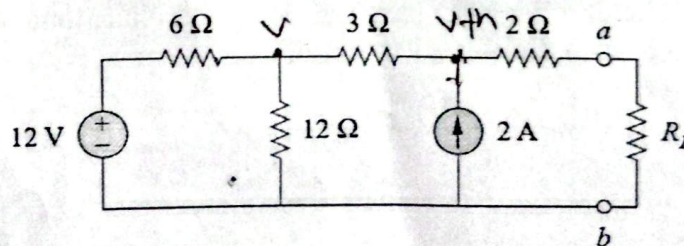


Fig. 5(c)

- Q6. (a) Derive the Source free RL circuit response with appropriate graphs for current. CLO2 [4]

- (b) Find I in the circuit of Fig. 6(b). CLO3 [6]

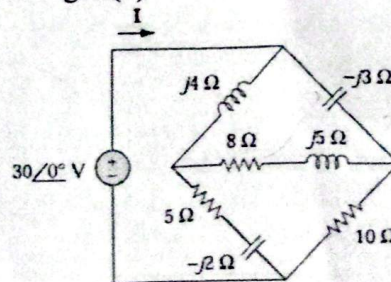


Fig. 6(b)



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

ENG 111: English Basics
Final Examination
Spring 2026

30th Batch

1st Semester

10 September 2026

Time: 02 Hours

Total Marks: 40

CLOs Questions/ Statements

Marks

CLO5 01. Rewrite the following paragraph using punctuation and capitalization: 05

Coding has become an important skill in everyday life, but many people still wonder why should students learn it. Teachers and technology experts believe that coding teaches students how to solve problems step by step and turn their ideas into practical solutions. Beginners often start with simple languages such as Python, Scratch or JavaScript, depending on their goals. Coding can also improve creativity, patience and logical thinking. When students write a program they may make mistakes but by finding and fixing those errors they gradually learn how to think more carefully and work more effectively.

CLO5 02. Fill in the blanks with suitable articles: 05

Scientific discoveries often begin with curiosity and careful observation. Marie Curie was (a) a remarkable scientist whose research on radioactivity opened new possibilities in medicine and physics. She became (b) the first woman to receive a Nobel Prize and remains one of (c) the most respected scientists in history. Albert Einstein was (d) a physicist whose theory of relativity changed our understanding of space and time. Later, he joined (e) a university in (f) the United States, where he continued his academic work. Alexander Fleming was (g) a Scottish scientist who discovered penicillin after observing the growth of bacteria in his laboratory. His discovery became (h) a major breakthrough in modern medicine. Today, (i) the scientific community continues to build on such discoveries, while (j) the contributions of these scientists inspire researchers around the world.

CLO5 03. Change the following sentences into different degrees of comparison: 05

- a) He is no less intelligent than you. (Positive)
- b) No other crime in human life is as dreadful as lying. (Comparative)
- c) My siblings are as wicked as my cousins. (Comparative)
- d) Very few scientists in Bangladesh are as promising as Dr. Ferdous. (Superlative)
- e) Alam's cousin is not smarter than him. (Positive)

... is no less intelligent as you.

CLO5 04. Transform the following sentences into indirect narration:

05

- a) The teacher said to the students, "Complete your assignment before tomorrow."
- b) Rafi said to me, "I have never visited Cox's Bazar."
- c) The citizens said, "May our country prosper and develop."
- d) The old man said to the boy, "Please help me cross the road."
- e) Maria said to her friend, "Did you attend the seminar yesterday?"

Human

CLO4 05. Write a paragraph describing the atmosphere of your university during a Cultural Festival.

10

CLO4 06. Write a resignation letter from your position as a Marketing Executive, citing limited professional growth as the reason of your resignation.

10



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Batch: 30th

Semester: Spring 2025

Date: 03/09/2026

Final Examination

Time: 2 hours

Total Marks: 40

Answer any five questions.

5 x 8 = 40

1. (a) Suppose that a connected planar simple graph has 30 vertices, each of degree 3. Into how many regions does a representation of this planar graph split the plane? CLO 4 2

- (b) State and explain the different procedures to determine Minimum Spanning Tree (MST) CLO 4 6

2. (a) Define Linearly Ordered Set and POSET. CLO 4 2

- (b) Let $A = \{2, 3, 6, 12, 24, 48, 96\}$ be a set. Find out $R = \{(a, b) \mid a \text{ divides } b\}$. Draw the partial ordering diagram for R and finalize the Hasse diagram from the partial ordering diagram, also find out the maximal and minimal. CLO 4 6

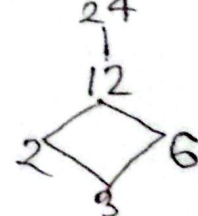
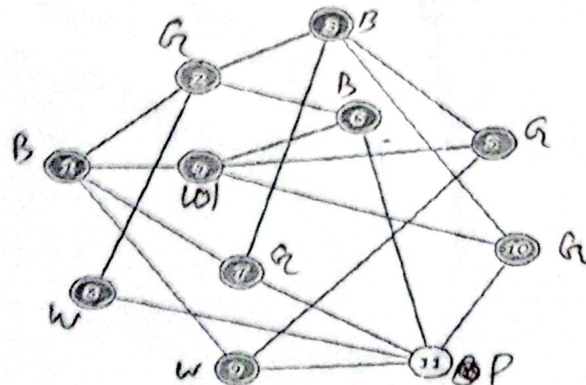
3. (a) State and explain the Handshaking theorem. CLO 4 2

- (b) Write the similarities and dissimilarities among adjacency matrix, adjacency list and adjacency multilist. CLO 4 6

4. (a) State the Dirac's theorem and Ore's theorem. CLO 4 2

- (b) Define Composite relation. Let $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$ and $C = \{0, 1, 2\}$ be 3 sets. R be a relation from A to B and S be a relation from B to C , where $R = \{(a, b) \mid a \leq b\}$ and $S = \{(b, c) \mid b \leq c\}$. Find R ; S ; $S \circ R$; $R \circ S$. CLO 4 6

5. (a) Color the graph by using graph coloring technique and find the chromatic number. CLO 4 3

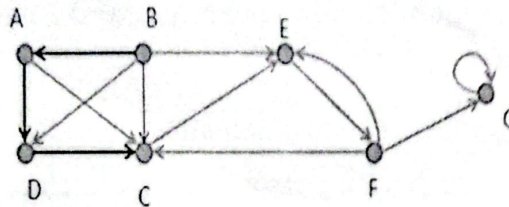


- (b) Let $A = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 5 & 4 \\ 2 & 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 5 & 4 \\ 2 & 3 & 2 \end{bmatrix}$ be 3×3 matrices. Find $A + B$, $A \times B$ and the inverse of the matrix B . CLO 4 5

6. (a) Prove that the set of even integers forms Abelian group. CLO 1 CLO 4 2

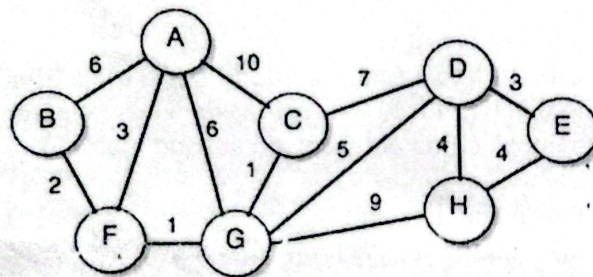
$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

- (b) Find the properties (with reason), adjacency matrix and adjacency list for the following di-graph: CLO 4 6

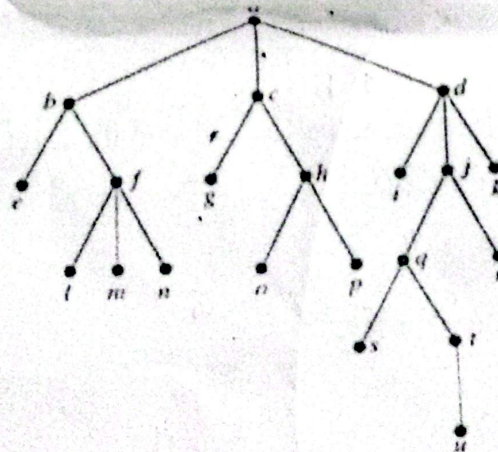


Determine degree, in-degree and out-degree of each vertex for the above graph.

- 7 (a) Find the minimum spanning tree for the following weighted graph by using (i) Prim's algorithm (ii) Kruskal's algorithm CLO 4 4



- (b) Briefly describe tree traversal techniques and find the order of the elements using tree traversal techniques after traversing the following tree: CLO 4 4





HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 117: Engineering Ethics and Cyber Law

Batch: 30th (Sec:1 & Sec:2)

Semester: Spring 2026

Date: 06/09/2026

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

1. (a) What is ransomware? How does it differ from phishing attacks? CLO5 2
(b) Define cybercrime and explain two main types with examples. CLO5 3
(c) Demonstrate the key challenges in establishing jurisdiction in cyberspace. CLO4 3

server

2. A cybercriminal sends fake emails pretending to be from a bank, tricking users into revealing their login credentials. Many victims lose their money as a result.
(a) What type of cybercrime does this scenario describe? Explain how it works. CLO5 4
(b) What preventive measures should individuals take to protect themselves from such attacks? CLO5 4

3. (a) What is copy right? Define the following terms in the context of copy right:
i. Publication
ii. Author
iii. Work CLO4 4
(b) Explain every process for determining the first owner of the work. CLO4 4

4. A popular social media influencer in Bangladesh has approached your software company with the following requests:
-To create multiple fake accounts using real people's photos and information.
-To develop a bot that automatically downloads and reposts viral content from other creators.
-To implement a system that artificially inflates view counts and engagement metrics.

- (a) Identify which specific provisions of the Digital Security Act 2018 and ICT Act 2006 would be violated by these requests CLO4 4
(b) As the lead engineer assigned to this project: CLO2 4
i. What ethical considerations should you weigh?
ii. How would you handle this situation while maintaining professional integrity?

5. (a) What is Firewall? Explain the types of Firewall. CLO2 4
(b) Explain DoS and DDoS Attack with proper example. CLO2 4

6. A university student gains unauthorized access to their academic grading system and changes their grades. The student later argues that they did not intend to cause harm and were only testing system security.

- (a) Is this act considered ethical under professional computing standards? CLO2 4
Defend your answer.
(b) How did the meaning and implications of the term "Hacking" evolve from the 1960s to the Web era? CLO4 4

7. (a) Define Intellectual Property (IP). What are its key types? CLO3 2
(b) Explain the differences between copyright, patent, trademark, and trade secret. CLO3 3
(c) How does the Bangladesh Copyright Act, 2000 protect the rights of content creators? CLO4 3



HAMDARD UNIVERSITY BANGLADESH

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Faculty of Science, Engineering and Technology

Course Code: MAT115

Course Title: Differential and Integral Calculus

Final Examination

Semester: Spring 2026

Date: 9/1/2026

Total Time: 2:00 Hours

Total Marks: 40

Answer any five (05) out of the following questions:

5 × 8 = 40

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No.	Questions	CLOs	Marks
Q1.	(a) Show that the area bounded by the circle $r = 2a \cos \theta$ is πa^2 .	CLO3	[2.5]
	(b) Evaluate the following multiple integration: $\int_0^{2\pi} \int_0^{\pi} \int_0^a r^2 \sin \theta \, dr \, d\theta \, d\phi$	CLO3	[2.5]
	(c) If $\mathcal{R}$ is a region bounded by $x = 1$, $x = 4$, $y = -1$ and $y = 2$, then determine $\iint (2x + 6x^2y) \, dy \, dx$.	CLO3	[3]
Q2.	(a) State and prove the Walli's theorem and hence determine the followings: (i) $\int_0^{\frac{\pi}{2}} \cos^9 x \, dx$ (ii) $\int_0^{\frac{\pi}{2}} \sin^{10} x \, dx$	CLO2 CLO3	[6]
	(b) Determine the following: $\lim_{n \rightarrow \infty} \left[\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \dots + \frac{n}{n^2+n^2} \right]$	CLO3	[2]
Q3.	(a) Define definite integral as the limit of a sum.	CLO1	[1]
	(b) State and prove the fundamental theorem of integral calculus.	CLO2	[5]
	(c) Evaluate the following by using the first principle of integration: $\int_a^b e^x \, dx$	CLO3	[2]
Q4.	(a) Show that the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $ab\pi$.	CLO2	[4]
	(b) Prove that $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} \, dx = \frac{\pi}{4}$	CLO3	[4]
Q5.	(a) Define Beta and Gamma functions.	CLO1	[1]
	(b) Derive the relation between Beta and Gamma functions.	CLO2	[4.5]
	(c) Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$	CLO3	[2.5]



HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

- ~~Q6.~~ (a) State and prove Chain rule for determining total differential coefficient.
 (b) If $u = \tan^{-1} \frac{x^3+y^3}{x-y}$, then prove the followings:

$$(i) x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

$$(ii) \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right)^2 u = (1 - 4 \sin^2 u) \sin 2u$$

- Q7. (a) Obtain the reduction formula for $I_n = \int \sec^n x \, dx$ and hence determine I_6 .

- (b) Prove that

$$\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}$$

and hence determine $\int_0^{\frac{\pi}{2}} \sin^8 x \cos^6 x \, dx$.

$\cos x + 1$
 $\tan^2 x = \sec^2 x - 1$
 $\sec x - 1 = \frac{d}{dx} (f)$

$$x \frac{d}{dx} (f) = \frac{d}{dx} (f) + y \frac{d}{dy} (f) = 1$$



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY 111

Final Examination

Date: 9/8/2026

Total Time: 2 Hours

Course Title: Physics I

Semester: Spring 2026

Total Marks: 40

Answer any four (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

	CLOs	Marks
Q1. (a) What is optical aberration?	CLO1	[1]
(b) What are the main types of optical aberration?	CLO1	[2]
(c) Explain spherical aberration with a neat diagram	CLO2	[4]
(d) Explain the concept of curvature of a lens with a suitable diagram.	CLO2	[3]
Q2. (a) Define astigmatism.	CLO1	[1]
(b) Explain wavelets with suitable figure.	CLO2	[2]
(c) State the Rayleigh criterion for resolution.	CLO1	[5]
(d) Why does diffraction take place?	CLO2	[2]
Q3. (a) What is an interferometer?	CLO1	[1]
(b) What are the effects caused by spherical aberration?	CLO2	[2]
(c) Explain Malus' law.	CLO2	[4]
(d) The intensity of light through a polarizer and analyzer is maximum when their principal planes are parallel. Through what angle the analyzing nicol must be rotated so that the intensity gets reduced to $\frac{1}{4}$ of the maximum value.	CLO3	[3]
Q4. (a) State the Kelvin-Planck statement of the second law of thermodynamics.	CLO1	[1]
(b) Define molecular specific heat.	CLO1	[1]
(c) Explain the phenomenon of diffraction from a circular aperture.	CLO2	[3]
(d) Explain the construction and working principle of the Fresnel biprism with a neat diagram.	CLO2	[5]
Q5. (a) Define thermodynamics.	CLO1	[1]
(b) What do you mean by thermal equilibrium?	CLO1	[1]
(c) Describe some practical examples of the first law of thermodynamics.	CLO2	[4]
(d) State and explain the first law of thermodynamics.	CLO2	[4]
Q6. (a) State Brewster's law.	CLO1	[1]
(b) Sketch a diagram demonstrating light polarization.	CLO2	[2]

$$I_0 = I_0 \cos^2 \theta$$

77



- (c) Show that at Brewster's angle, the reflected and refracted rays are perpendicular to each other. CLO2 [4]
- (d) The refractive index for plastic is 1.25. Calculate the angle of refraction for a ray of light incident at polarizing angle. CLO3 [3]

$$P + R = 90^\circ$$
$$P = 90^\circ \times 1.25$$